INCH-POUND
MIL-PRF-3098/88J
w/AMENDMENT 1
16 August 2023
SUPERSEDING
MIL-PRF-3098/88J
1 November 2017

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR112/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 5 MHz to 20 MHz; fundamental; noncontrolled; series resonance.

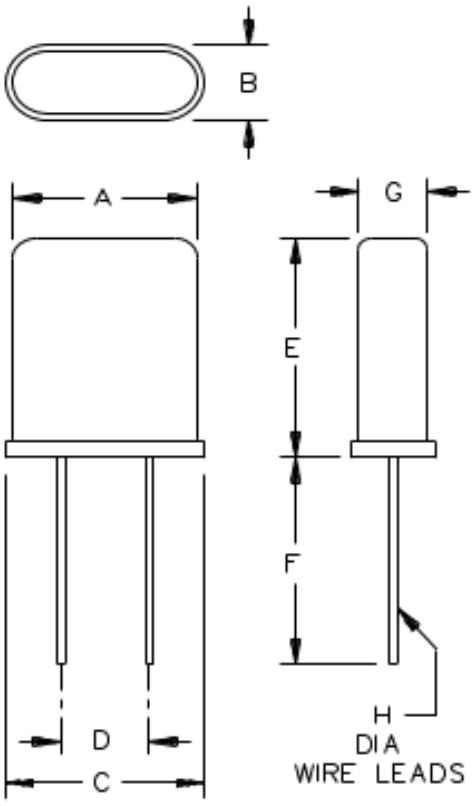


FIGURE 1. Crystal unit - CR112/U.

AMSC N/A
FSC 5955
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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.402	---	10.21
B	---	.183	---	4.65
C	---	.435	---	11.05
D	.184	.200	4.67	5.08
E	---	.530	---	13.46
F	.50	---	12.7	---
G	---	.150	---	3.81
H	.015	.019	0.38	0.48

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Type CR112/U, crystal unit - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 5 MHz to 20 MHz, inclusive.

Capacitance, shunt: 7 pF, maximum.

Rated drive level: 1.0 mW, maximum.

Frequency tolerance:

Operating temperature range: ±25 ppm.

Equivalent resistance:

Frequency range, inclusive	Maximum resistance
MHz	Ohms
5 to 7	50
7+ to 10	30
10+ to 15	25
15+ to 20	20

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: MIL-STD-202-201.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: This crystal unit is identical to CR60A/U, except for the tighter frequency tolerance of ±25 ppm.

Qualification: Not required. First article testing (FAT) shall be performed according to the qualification table II of MIL-PRF-3098; for product level B, Table II is applicable and for product level S, Table II-A is applicable. Thirty samples for product level B and forty-five samples for product level S shall be selected. Approval of FAT shall be at the discretion of the acquiring activity.

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Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-201

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
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Navy - EC
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DLA - CC

(Project 5955-2023-003)

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